

Sai Pramodh Varma Byrraju

[LinkedIn](#) | [Portfolio](#)

EDUCATION

Michigan State University | Honors College

December 2024

- B.S | Computer Science, Minor in Business, Computational mathematics science engineering (data science)
- Honors College | Dean's list

GPA 3.69

Notable Courses: CSE 498 (Capstone), CSE 404 (Intro to Machine Learning) CSE 335 (Object-Oriented Software Design, C++), CSE 480 (Data base Systems, C++), CSE 476 (Mobile Application Development, Java), CSE 331 (Algorithms and Data Structures , Python), CSE 440 (Intro to AI)

SKILLS

Programming Languages / OS

Python, C++, C, C#, Java, Typescript, SQL, Dart, R, Shell, HTML, CSS, Linux

Frameworks / Tools

Node.js, Angular, React.js, MongoDB, Git, FastAPI, Pandas, TensorFlow, PyTorch, Azure

Certifications

Agile Development (LinkedIn), CE in Software Development (Microsoft)

PROFESSIONAL EXPERIENCE

IT Intern | Tech Mahindra (Jackson National)

June 2024 – Aug 2024

- Managed **Power BI** dashboards and prepared cloud-based datasets, delivering insights for business units.
- Maintained the Enterprise Data Platform in **Azure** cloud environment.
- Resolved **25+ critical technical issues** in collaboration with IT teams optimizing **data-driven operations**, enhancing service delivery.

Capstone Application Developer | Whirlpool

Jan 2024 – May 2024

- Developed a **smart oven API** using **BERT**, now being integrated into Whirlpool's ovens, delivering personalized recipe recommendations based on user cooking habits
- Built a **full stack** application with Dart ensuring cross compatibility on Android and iOS , and a data-driven dashboard for data visualization using JavaScript and Plotly.
- Enhanced user experience by continuous integration and fine-tuning the ML model, leveraging **MongoDB** and **FastAPI**.
- Conducted test driven development with users, achieving **92% accuracy** in recipe recommendations based on 100+ user tests.

Student Accountant | MSU Business and Financial Services (BFS)

June 2022 – Jan 2023

- Displayed **problem-solving** skills through accuracy and thoroughness in tasks such as computing, data entry, and filing for **financial markets**.
- Handled several accounting necessities like operating statements, and reconciliations.

Teaching Assistant & Tutoring | MSU

Aug 2021 – Jan 2023

- Assisted students in courses as a Teaching assistant, and Tutor: CSE 335, CSE 232, CSE 102, ACC 230, EC 251, MTH 103
- Lead classes with over **70 students**. Provided 1:1 sessions, mentoring over **15 students** to learn and achieve a perfect 4.0 GPA.

Service Center Representative | MSU

May 2023 – Dec 2024

- Provided customer service and clerical support to residence hall students and visitors, managing inquiries.
- Handled mail, packages, and reception duties, maintaining a welcoming reception.

PROJECTS

DebugMore – Job Preparation Platform (React, Node.js, Python)

Dec 2024

- Developing an advanced platform using React, Node.js, and Python (Flask) to extract key skills from job descriptions and recommend coding problems to excel in interviews leveraging **Hugging Face Transformers** for high-accuracy **NER**.
- Designed and implemented a real-time multi-language code execution engine integrated with the **Piston API**, supporting Python, C++, Java, and C, along with a dynamic code generation feature for job-specific skill prep.
- Built a modular backend architecture with **RESTful APIs**, asynchronous processing, and robust **CORS handling**.

Custom Python DBMS (Python, SQL)

Oct 2024

- Created a **scalable Python module** emulating **SQLite3** functionality, enabling advanced SQL query and supporting complex data analysis pipelines for real-world database applications.

Smart Grocery Manager (Java, Android)

Oct 2023

- Implemented a grocery management application using Java, leveraging **object-oriented** principles for scalable code.
- Optimized application performance by managing activity lifecycles efficiently, utilizing intent-based transitions, and **memory management** to ensure smooth navigation.

INVOLVEMENTS

Association for Computing Machinery (ACM) | AI Project Manager | Michigan State University

- Involved in frequent meetings of the association to meet like-minded individuals that are passionate about computer science.
- Managed teams aiding them with technical interview skills and competitive programming.

Spartan Blockchain Solutions (SBS) | Technical Analyst | Michigan State University

- Committed with the non-profit blockchain firm in advising local businesses with innovative blockchain solutions.